專案 1 ： 三層 IP 網路架構 (Route-Access) 與 Rocky Linux 及 ISO 27001:2022 安全規範整合實作

平台： Cisco IOS/ GNS3 / Rocky Linux

安全框架： ISO 27001:2022

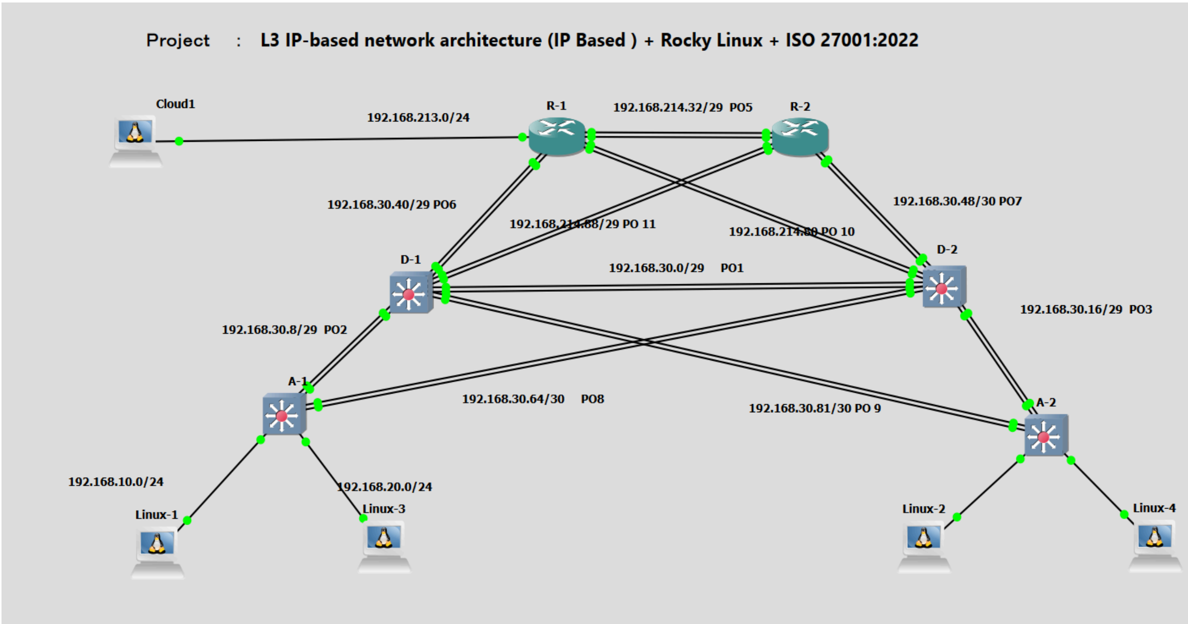
關鍵技術： L3 Routed Access, EtherChannel, Syslog, DHCP , HSRP, PAT, NTP, OSPF, Stratis File System (auto mount), ISO 27001:2022 Control Implementation

傳統問題： 在金融、醫療、中大型企業數據中心或總部網路 ， 傳統 Layer 2 架構依賴 STP 收斂速度慢、易產生廣播風暴，且缺乏針對無 VLAN 環境的資安隔離與合規管控

- **高可用性與擴充性 (Availability & Scalability)：** 全網採用全三層路由（Routed Core & Access）結合 Port-Channel 多線路冗餘，適合在要求零中斷服務企業環境中。此架構採用 Routed Access (L3 路由到接入層**)**，透過 OSPF 搭配 ECMP實現毫秒級收斂與流量負載分流 ， 未來新增機架 (Rack) 或樓層，只需分配新的 IP Subnet 並發布路由，不影響原有網路拓撲 。配合 Rocky Linux 建立集中式服務（Syslog 遠端日誌伺服器、NTP 時間同步伺服器、DHCP）與 Stratis 存取，能有效降低營運維護成本並符合系統 hardening（強化安全）要求。

- **ISO 27001:2022 安全框架控制 (Security Controls)：**「ISO 27001:2022 安全控制措施（Security Controls）：A.5.17 身分驗證資訊（強制執行使用者密碼生命週期管理與定期輪調、A.8.2 特權存取權（防止未授權的 Root 權限存取）、A.8.5 安全性認證（取消 Root 外部直接登入管理，配備 SSHv2 存取權、A.8.15日誌留存（Rocky Linux與L3設備集中系統日誌審計）、A.8.17時脈同步（確保整體網路架構與伺服器環境系統時脈一致）、A.8.20網路安全性（僅保留必要開放端口，並包圍不必要的網路服務與高風險協議）及A.8.24密碼學的使用（保護網路傳輸與管理界面之資訊機密性），以確保整體架構符合國際資安標準。

Logical Topology



IP Address Table

Device	Port	IP	Mask	Description
R1	g1	192.168.213.12	/24	connect to Gateway
	g2	shut down		
	g3	po5		connect to R-2
	g4	po5		connect to R-2
	g5	po6		connect to D-1
	g6	po6		connect to D-1
	g7	po6		connect to D-2
	g8	po6		connect to D-2
	po 5	192.168.30.33	/29	connect to R-2
	po 6	192.168.30.42	/29	connect to D-1
	po 10	192.168.30.81	/29	connect to D-2
	standby 5	192.168.30.35		connect to R-2 (HSRPv2)
	standby 6	192.168.30.46		connect to D-1 (HSRPv2)
	standby 10	192.168.30.86		connect to D-2 (HSRPv2)

D-2	g0/0	shut down		
	g0/1	shut down		
	g0/2	po1		connect to D-1
	g0/3	po1		connect to D-1
	g1/0	po3		connect to A-2
	g1/1	po3		connect to A-2
	g1/2	po7		connect to R-2
	g1/3	po7		connect to R-2
	g2/0	po8		connect to A-1
	g2/1	po8		connect to A-1
	g2/2	po10		connect to R-1
	g2/3	po10		connect to R-1
	g3/0	shut down		
	g3/1	shut down		
	g3/2	shut down		
	g3/3	shut down		
	po 1	192.168.30.2	/29	connect to D-1
	po 3	192.168.30.18	/29	connect to A-2
	po 7	192.168.30.42	/29	connect to R-2
	po 8	192.168.30.52	/29	connect to A-1

po 10	192.168.30.82	/29	connect to R-1
standby 1	192.168.30.6		connect to D-1 (HSRPv2)
standby 3	192.168.30.22		connect to A-2 (HSRPv2)
standby 7	192.168.30.46		connect to R-2 (HSRPv2)
standby 8	192.168.30.52		connect to A-1 (HSRPv2)
standby 10	192.168.30.86		connect to R-1 (HSRPv2)

DHCP Addresses are released to Linux-1 and Linux-2

```
root@localhost:/var/log/remote/127.0.0.1# cat 2026-08-08.log | grep DHCP4_LEASE_ALLOC
print $1, $2, $3, $4, $10,$16,$19,$20,$21,$22 }'
Aug 8 11:04:52 localhost DHCP4_LEASE_ALLOC 192.168.10.100 allocated for 7200 seconds
Aug 8 11:14:47 localhost DHCP4_LEASE_ALLOC 192.168.20.110 allocated for 7200 seconds
Aug 8 11:15:02 localhost DHCP4_LEASE_ALLOC 192.168.20.110 allocated for 7200 seconds
```

```
A-1#sh run int g2/0
Building configuration...

Current configuration : 206 bytes
!
interface GigabitEthernet2/0
 no switchport
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.213.14
 ip ospf authentication key-chain OSPF-KEY
 ip ospf 100 area 0
 negotiation auto
end
```

```
[root@rocky-cloud ~]# nmcli dev show | grep IP4
IP4.ADDRESS[1]:                         192.168.10.100/24
IP4.GATEWAY:                            192.168.10.1
IP4.ROUTE[1]:                           dst = 192.168.10.0/2
IP4.ROUTE[2]:                           dst = 0.0.0.0/0, nh
IP4.DNS[1]:                             8.8.8.8
IP4.ADDRESS[1]:                         127.0.0.1/8
IP4.GATEWAY:                            --
[root@rocky-cloud ~]#
```

```
[root@rocky-cloud ~]# nmcli con show
NAME  UUID                                  TYPE      DEVICE
ens3  fbc49833-4cdb-548b-a293-c38b39836fdb  ethernet  ens3
lo    176310ca-23f3-40a8-b3a7-c51870b95278  loopback  lo
[root@rocky-cloud ~]# nmcli dev show | grep IP4
IP4.ADDRESS[1]:                         192.168.20.110/24
IP4.GATEWAY:                            192.168.20.1
IP4.ROUTE[1]:                           dst = 192.168.20.0/24, nh = 0.0.0.0, mt = 100
IP4.ROUTE[2]:                           dst = 0.0.0.0/0, nh = 192.168.20.1, mt = 100
IP4.DNS[1]:                             8.8.8.8
IP4.ADDRESS[1]:                         127.0.0.1/8
```

Centralize Log Server

```
root@localhost:/var/log/remote# ls
127.0.0.1  192.168.213.12  192.168.30.1  192.168.30.11  192.168.30.19  192.168.30.75
root@localhost:/var/log/remote# cat 192.168.30.1
cat: 192.168.30.1: Is a directory
root@localhost:/var/log/remote# cd 192.168.30.1
root@localhost:/var/log/remote/192.168.30.1# ls
2026-08-08.log
root@localhost:/var/log/remote/192.168.30.1# cat 2026-08-08.log
Aug  8 20:12:37 192.168.30.1 139: *Aug  8 10:05:36.050: %OSPF-5-ADJCHG: Process 100, Nbr 192.1
68.215.1 on Port-channel11 from LOADING to FULL, Loading Done
```

```
A-2(config)#logg
A-2(config)#logging on
A-2(config)#logging host 192.168.213.14
A-2(config)#logging h
*Aug  8 06:07:13.164: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.213.14 port 514 started - CLI initiate
% Incomplete command.

A-2(config)#logging tr
A-2(config)#logging trap 5
```

PAT (Port Address Translate)

```
R1#
R1#show ip nat translations
Pro  Inside global         Inside local          Outside local         Outside global
tcp  192.168.213.12:5068   192.168.10.100:33732  34.49.197.38:443      34.49.197.38:443
tcp  192.168.213.12:5067   192.168.10.100:33722  34.49.197.38:443      34.49.197.38:443
tcp  192.168.213.12:5062   192.168.20.110:56328  34.49.197.38:443      34.49.197.38:443
tcp  192.168.213.12:5069   192.168.10.100:33742  34.49.197.38:443      34.49.197.38:443
Total number of translations: 4
```

HSRP (Hot Standby Routing Protocol)

```
D-2#sh  standby br
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Po1         1    100   Standby 192.168.30.1    local           192.168.30.6
Po3         3    110 P Active  local           192.168.30.18   192.168.30.22
Po7         7    100   Standby 192.168.30.41   local           192.168.30.46
Po8         8    110 P Active  local           192.168.30.50   192.168.30.52
Po10        10   100   Standby 192.168.30.81   local           192.168.30.86
```

NTP (Network Time Protocol)

```
#
# add network in /etc/chrony.conf
allow 192.168.213.0/24
allow 192.168.30.0/24
```

```
root@localhost:~# chronyc sources -v

  .-- Source mode  '^' = server, '=' = peer, '#' = local clock.
 / .- Source state '*' = current best, '+' = combined, '-' = not combined,
| /             'x' = may be in error, '~' = too variable, '?' = unusable.
||                                                 .- xxxx [ yyyy ] +/- zzzz
||      Reachability register (octal) -.           |  xxxx = adjusted offset,
||      Log2(Polling interval) --.      |          |  yyyy = measured offset,
||                                \     |          |  zzzz = estimated error.
||                                 |    |           \
MS Name/IP address         Stratum Poll Reach LastRx Last sample
===============================================================================
^+ 103.186.118.217               2   6    17    32   +593us[ +593us] +/- 8188us
^- 123-204-232-128.adsl.dyn>     2   6    17    32  +1365us[+1365us] +/-   56ms
^- 103.186.118.222               2   6    17    32   -475us[ -475us] +/-   13ms
^* 103.186.118.212               2   6    17    32   -172us[-1553us] +/- 9388us
```

```
D-1#sh ntp associations

  address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.213.14  103.186.118.212  3     22     64     3  7.205 4761.23 1937.8
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

OSPF-Neighborship with Authentication

```
D-2#sh ip  os neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.213.12    1   FULL/DR         00:00:38    192.168.30.81   Port-channel10
192.168.30.89     1   FULL/DR         00:00:38    192.168.30.41   Port-channel7
192.168.30.75     1   FULL/BDR        00:00:35    192.168.30.18   Port-channel3
192.168.30.90     1   FULL/DR         00:00:38    192.168.30.1    Port-channel1
D-2#
```

```
key chain OSPF-KEY
 key 200
   key-string 7 0110050A5A
   cryptographic-algorithm hmac-sha-256
password encryption aes
```

```
interface Port-channel3
 no switchport
 ip address 192.168.30.19 255.255.255.248
 standby version 2
 standby 3 ip 192.168.30.22
 standby 3 priority 110
 standby 3 preempt
 ip ospf authentication key-chain OSPF-KEY
 ip ospf 100 area 0
```

Stratis File System : auto mount to client

```
root@localhost:~# stratis filesystem list
Pool    Filesystem    Total / Used / Free / Limit             Device                          UUID
pool1   fs1           1 TiB / 546 MiB / 1023.47 GiB / None    /dev/stratis/pool1/fs1          3dac3735-2032-4702-b249-32cca0293a46
pool1   snapshot1     1 TiB / 546 MiB / 1023.47 GiB / None    /dev/stratis/pool1/snapshot1    41202873-6a39-474a-802a-ed24550098dd
```

```
[root@rocky-cloud ~]# df -h
Filesystem                       Size  Used Avail Use% Mounted on
/dev/vda4                        8.4G  1.1G  7.4G  13% /
devtmpfs                         594M     0  594M   0% /dev
tmpfs                            633M     0  633M   0% /dev/shm
tmpfs                            254M  552K  253M   1% /run
tmpfs                            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
/dev/vda3                        936M  279M  658M  30% /boot
/dev/vda2                        200M   14M  186M   7% /boot/efi
tmpfs                            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs                            1.0M     0  1.0M   0% /run/credentials/serial-getty@ttyS0.service
tmpfs                            127M  4.0K  127M   1% /run/user/1000
192.168.213.14:/srv/stratis_share  17G  5.4G   12G  32% /mnt/client_stratis
```

Comply with ISO 27001:2022 - A.8.2—特權存取權限 (要求對特權存取權限的分配和使用進行嚴格控制和限制)，A.8.5—安全認證 (緩解針對管理員帳戶的暴力破解攻擊)

```
root@localhost:~# grep -i "PermitRootLogin" /etc/ssh/sshd_config
#PermitRootLogin prohibit-password
# the setting of "PermitRootLogin prohibit-password".
PermitRootLogin no
```

Comply with ISO 27001:2022： A.8.20—網路安全 (最少的開放網路端口) , A.8.9— 配置管理 (服務最小化)

```
root@localhost:~# firewall-cmd --list-services
cockpit dhcp dhcpv6-client mountd nfs ntp rpc-bind ssh
```

Comply with ISO 27001:2022 ：A.5.17—身份驗證訊息 (強制執行使用者密碼生命週期管理和定期輪調), A.8.9 (強制執行標準化的 Linux 帳戶老化和密碼原則基準)

```
root@localhost:~# chage --maxdays 60 peter
root@localhost:~# chage --list peter
Last password change                                    : Aug 09, 2026
Password expires                                        : Oct 08, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 60
Number of days of warning before password expires       : 7
```

Comply with ISO 27001:2022 : A.8.15—日誌記錄 (產生、匯總和儲存遠端事件日誌，以便進行分析和稽核)

```
root@localhost:/var/log/remote# ls
127.0.0.1  192.168.213.12  192.168.30.1  192.168.30.11  192.168.30.18  192.168.30.19  192.168.30.75
root@localhost:/var/log/remote# cat 192.168.30.11
cat: 192.168.30.11: Is a directory
root@localhost:/var/log/remote# cd 192.168.30.11
root@localhost:/var/log/remote/192.168.30.11# ls
2026-08-08.log  2026-08-09.log
root@localhost:/var/log/remote/192.168.30.11# cat 2026-08-09.log
Aug  9 14:11:39 192.168.30.11 93: *Aug  9 06:11:25.889: %HSRP-5-STATECHANGE: Port-channel8 Grp 8 state Sp
eak -> Standby
Aug  9 14:12:01 192.168.30.11 94: *Aug  9 06:11:47.447: %PLATFORM-5-SIGNATURE_VERIFIED: Image 'flash0:/vi
os_l2-adventerprisek9-m' passed code signing verification
Aug  9 15:41:42 192.168.30.11 98: *Aug  9 07:40:17.045: %SYS-5-CONFIG_I: Configured from console by conso
le
Aug  9 15:44:25 192.168.30.11 101: *Aug  9 07:43:00.584: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was w
ritten to disk successfully.
root@localhost:/var/log/remote/192.168.30.11# S
```

Comply with ISO 27001:2022 : A.8.17—時鐘同步 (確保整個環境中的系統時鐘與標準時間源保持一致)

```
R-2#sh ntp associations

  address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.213.14  103.186.118.215  3    259    256   377  1.799 1116404  3.329
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

```
D-1#sh ntp associations

  address         ref clock       st   when   poll reach  delay  offset   disp
*~192.168.213.14  103.186.118.212  3     22     64     3  7.205 4761.23 1937.8
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
```

Comply with ISO 27001:2022 : A.8.20 網路安全 (關閉阻止不需要的網路服務)

```
no cdp run
```

```
no ip http server
no ip http secure-server
```

```
no ip source-route
```

Comply with 27001:2022 - A.8.24 密碼學的使用 (有效地使用加密技術進行控制，以保護資訊的機密性、真實性和完整性。)